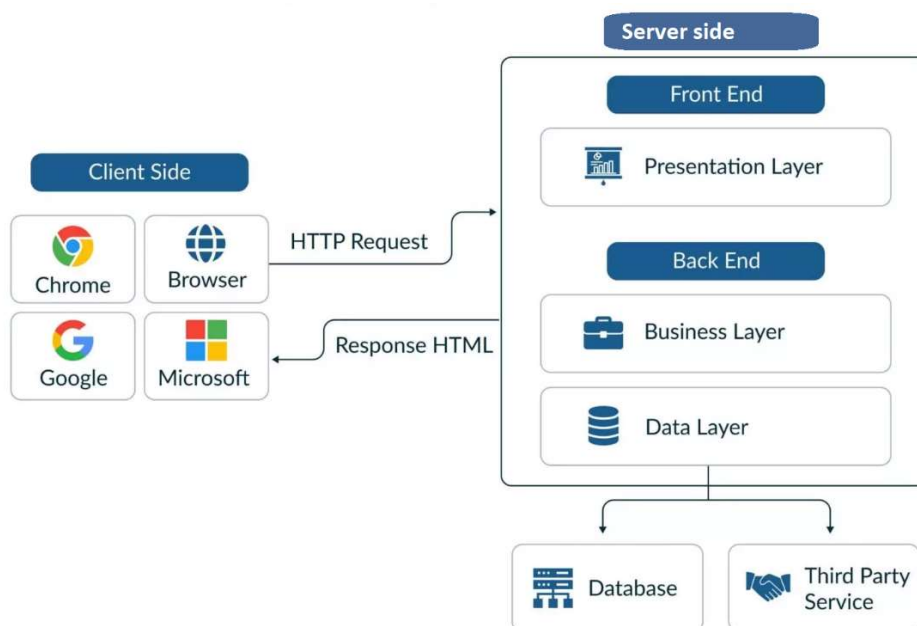


1 Definitions & History:

1.1 The WEB

The term "Web" (English name meaning "web"), a contraction of "World Wide Web" (hence the acronym www), is one of the possibilities offered by the Internet network to navigate between documents linked by hypertext links. The concept of the Web was developed at CERN (European Organization for Nuclear Research) in 1991 by a team of researchers that included Tim-Berners LEE, the creator of the hyperlink concept, considered today as the founding father of the Web. Tim Berners-Lee created the World Wide Web Consortium (W3C) which defines new versions of Web-related languages. More recently, he also created the World Wide Web Foundation, which analyzes and monitors the evolution of the Web. The real revolution of the Web lies in its universal character, the result of the use of recognized standards which are essentially:

- HTML, for the description of the pages available on the Web;
- HTTP, for communication between browser and web server;
- TCP/IP, the widely used network protocol;
- CGI, the interface that allows remote processing to be triggered on Web servers. The principle of the web is based on the use of hyperlinks to navigate between documents (called "web pages") using a software called a browser.



1.2 Internet:

Internet is the global computer network available to the public. It is a network of networks, made up of millions of networks, both public and private. Information is transmitted over the Internet through a standardized set of data transfer protocols.

1.3 Internet & WEB

Don't confuse the Internet and the Web, the Web is part of the Internet. The Internet is a large entity that includes, among other things: the Web, e-mail, instant messaging, etc. Tim Berners-Lee is therefore not the inventor of the Internet, he is only the inventor of the Web.

1.4 URL (Uniform Resource Locator):

On the Internet, documents are identified by a unique address, called a **URL**, which allows a resource to be located on any server on the Internet. A **URL** is a universal naming format for a resource on the Internet. It is a string of characters that is broken down into five parts:

- The name of the protocol: The language used to communicate on the network. The most widely used protocol is the HTTP protocol, the protocol for exchanging web pages in HTML format.
- Username and password: allows you to specify the access parameters to a secure server. This option is not recommended because the password is visible in the URL
- The server name: This is a domain name of the computer hosting the requested resource. By convention, web servers have a name starting with www., in order to symbolize that they are servers dedicated to the use of the web
- Port number: Number associated with a service that allows the server to know what type of resource is requested. The default port associated with the protocol is port number 80. Thus, when the server's Web service is associated with port number 80, the port number is optional
- Resource path: Allows you to specify the location of the document on the machine, i.e. the location (directory) and the name of the requested file.

A URL therefore has the following structure: <http://www.monsite.dz/test/test-url.html>

Protocol	Password (optional)	Server name	Port (optional if 80)	Path
http://	User:password@	www.monsite.dz	:80	/test/test-url.html

The file name in the URL can be followed by a question mark and then by data, this is additional data sent as a parameter of an application on the server.

The URL will then look like a string like this: `http://www.monsite.dz/test/test-url.html?toto=1&tutu=2`

The following protocols can be used via the URL, for example:

- ❖ http, for viewing web pages
- ❖ ftp, for viewing FTP sites
- ❖ telnet, for connecting to a remote terminal
- ❖ mailto, for sending an e-mail

1.5 Web server

Web server is software that allows clients to access web pages from a browser installed on their remote computer. To be accessible, a website must be published on a server called a web server. A web server embeds and runs server software (Apache, Microsoft IIS, nginx, etc.). Several Web servers exist:

- Apache Windows/ UNIX (free) (www.apache.org)
- Microsoft IIS (Internet Information Services): Microsoft's paid Web server
- Microsoft PWS (Personal Web Server)
- Iplanet Web Server: Netscape's paid server
- Etc,...

1.6 Web client

The Web client is software which is most often a web browser. The **browser** is capable of querying web servers, analyzing the code of web pages (HTML, CSS, javascript, etc.), exploiting their results and producing a visual result that is easy for a human to read.

The main browsers used on the Internet are: Mozilla Firefox, Microsoft Internet Explorer, Chrome, and Safari.

1.7 HTTP Protocol

Beyond links connecting formatted documents, the web takes on its full meaning with the HTTP protocol, which makes it possible to link documents hosted by remote computers.

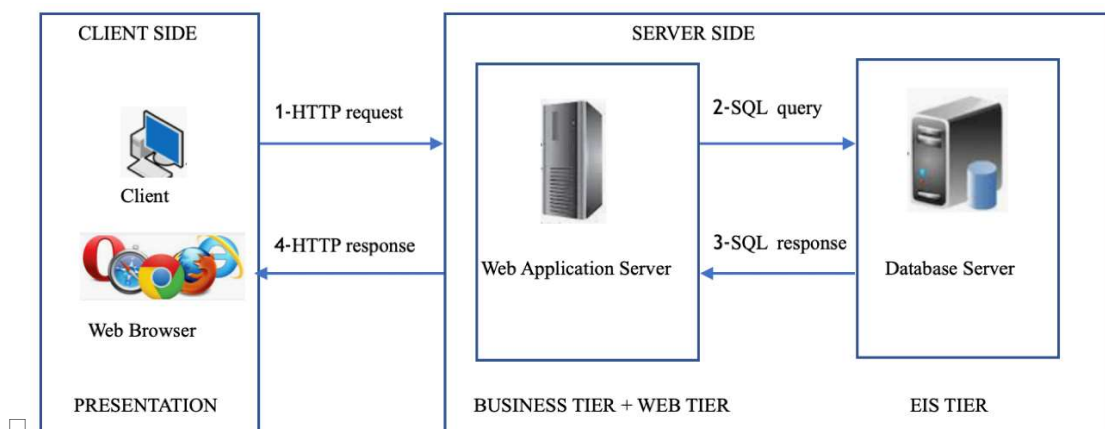
HTTP (HyperText Transfer Protocol), a web transfer protocol, its invention in 1989 at CERN in Geneva, at the same time as the HTML language, led to the appearance of the World Wide Web. HTTP has been the most widely used protocol since 1990.

The purpose of the protocol is to allow a transfer of files (mainly in HTML format) between a browser (the client) and a web server (Apache for example). The files are located through a URL between a browser (the client) and a web server.

Versions:

- **Version 0.9:** very basic version, only a GET command to retrieve a document, without receiving any information about its type.
- **Version 1.0: 1996**, allows messages with headers describing the content of the message to be forwarded using MIME (Multipurpose Internet Mail Extensions) encoding.
- **Version 1.1: 1997**, some improvements, including better cache management
- **Version 2:** very recent (2015), little used

1.7.1 Communication between Client and Web Server



- 1) User enters the **URL** (Uniform Resource Locator) of the website or file. The Browser then requests the **DNS** (DOMAIN NAME SYSTEM) Server.
- 2) **DNS Server** lookup for the address of the **WEB Server**.
- 3) The **DNS Server** responds with the **IP address** of the **WEB Server**.
- 4) The Browser sends over an **HTTP/HTTPS** request to the **WEB Server's IP** (provided by the **DNS server**).
- 5) The Server sends over the necessary files of the website to the browser.
- 6) The Browser then renders the files and the website is displayed. This rendering is done with the help of **DOM** (Document Object Model) interpreter, **CSS** interpreter, and **JS Engine** communally known as the **JIT** or (Just in Time) Compilers.
- 7) The browser requests the objects embedded in the server and displays the correctly formatted page;
- 8) To display the page, it is based on:
 - The default values of the browser,
 - The user's preferences set in the browser,
 - The values set in the document or style sheets. Examples: color of links (visited or not), text, page background, fonts, etc.

1.7.2 HTTP request:

A set of lines sent to the server by the browser. It includes:

- A request line: this is a line specifying the method that should be applied, the URL (the address of the page on the server) and the version of the protocol used.
- Request header fields: A set of optional lines to give additional information about the request and/or the client (Browser, operating system, etc.). Each of these lines is composed of a name qualifying the type of header, followed by a colon (:) and the value of the header.
- The request body: A set of optional lines that must be separated from the previous lines by an empty line and allows, for example, a data to be sent by a POST command when data is sent to the server by a form.

1.7.3 HTTP commands:

Command	Description
GET	Request the resource located at the specified URL (optionally given by URL with ?a=b&...)
HEAD	Request only the header of the resource located at the specified URL
POST	Send data to the program located at the specified URL
PUT	Place the resource at the specified URL (in practice rarely used)
DELETE	Delete the resource located at the specified URL (in practice rarely used)

HTTP request format:

Command, URL, protocole version <crLf>

HEADER : Value<crLf>

...

HEADER: Value<crLf>

Blank row <crLf>

REQUEST BODY

Example :

GET /rep/page.html HTTP/1.1

Host: serv.com

Accept: text/html

User-Agent: Mozilla/5.0 (Macintosh)

To complete the query, the request body is sent. It can contain, for example, the content of an HTML form sent as a POST.

1.7.4 HTTP response:

A set of lines sent to the browser by the server. It includes:

- A status line: this is a line that includes three elements that must be separated by a space: the version of the protocol used and the status of the processing of the request using a code and an explanatory text.
- Response header fields: this is a set of optional lines to provide additional information about the response and/or server. Each of these lines consists of a name qualifying the type of header, followed by a colon (:) and the value of the header
- The body of the response: it contains the requested document.

1.7.5 HTTP response codes:

Code	Message	Description
10x	Informational message	Codes not used in versions 1.0/1.1 of the protocol
20x	Success	The transaction went well (page found...)
30x	Redirect	The resource is no longer in the indicated location
40x	Error due to the client	The request was incorrect
403	FORBIDDEN	Access to the resource forbidden
404	NOT FOUND	The server did not find the resource at the specified address
50x	Error due to the server	An internal error occurred to the server

HTTP response format:

```
VERSION-HTTP CODE EXPLANATION<crLf>
HEADER: Value<crLf>
...
HEADER: Value<crLf>
Blank row<crLf>
RESPONSE BODY
```

Example:

```
HTTP/1.0 200 OK
Date : Sat, 15 Jan 2000 14:37:12 GMT
Server : Microsoft-IIS/2.0
Content-Type : text/HTML
<html>
<!-- The HTML code of the page -->
</html>
```

1.7.6 WEB 2.0

Principle of the classic Web

- ✓ One-to-many communication (= dissemination);
- ✓ Organizations that own sites that decide what information is on them.

Web 2.0 Principle

- ✓ Give control of information to users;
- ✓ To bring out social networks;
- ✓ Anyone can upload content to:
 - Give their opinion on a given subject (blogs)
 - Share their documents (images, videos, etc.)
 - Tag existing content

Technical aspects

- ✓ Tools (protocols, clients, servers) identical to those of the classic Web;
- ✓ Requires more computing power (dynamic pages);
- ✓ Requires more storage space (user-submitted content).

Examples of Web 2.0 sites:

Wikipedia, Dailymotion, YouTube....